NYAABA ADALEBSE TURIBIUS

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Product Design Engineer| Digital Engineering| CAD • CAE • Simulation

Biomedical Engineer with a strong focus on product design, digital engineering, simulation and technology-driven problem solving. Experienced in transforming engineering concepts into functional and manufacturable solutions through CAD, CAE and virtual testing. My engineering journey has developed from designing medical devices to applying digital engineering to real-world problems, including leading an 8-member team in the Siemens Immersive Design Challenge to develop a virtual prosthetics testing environment for children and adults.

EDUCATION

BSc. Biomedical Engineering
Kwame Nkrumah University of Science and Technology (KNUST) | 2023-2026

Biomedical Engineering Students' Association, KNUST Head of trade and Technology (Tratech)

- Led technical development initiatives that have trained 300+ biomedical engineering students in practical engineering skills.
- Design and coordinate workshops and practical sessions covering CAD, CAE, CFD, programming, electronics and product design.
- Help students transition from theoretical classroom knowledge to practical engineering application through hands-on projects and technical training.
- Led initiatives that strengthen the connection between academia, engineering practice, healthcare and industry.
- Mentor students in engineering tools, technical problem solving and product-development thinking.
- Promote a culture of practical innovation by encouraging students to develop solutions to real-world engineering and healthcare challenges

RELEVANT EXPERIENCE

DIPPER Lab (Distributed IoT-Platforms, Privacy and Edge-Intelligence Research) | September 2025 – August 2026

- Lead 3D modelling, product design, and simulation activities across multiple biomedical engineering projects, translating concepts into detailed and manufacturable designs.
- Contribute to the development of five key medical-device projects: High-Intensity Phototherapy Device, Sleep Apnea Monitor, Ezytherm, Neonatal Incubator and Automated Neonatal Device (ANET).
- Apply CAD and CAE approaches to refine product geometry, evaluate design performance and support engineering decisions before physical prototyping.
- Work across the product-development process from concept development and 3D modelling to simulation, design refinement and manufacturability considerations.
- Collaborate with multidisciplinary team members to balance functionality, performance, usability and manufacturing requirements

SIEMENS Team Lead | Siemens Immersive Design Challenge

- Led an 8-member engineering team in the development of a Virtual Prosthetics Test Lab using digital engineering and simulation concepts.
- Designed and evaluated different prosthetic concepts digitally for children and adults, reducing the dependence on repeated physical iterations during the early design stage.
- Applied virtual testing to investigate structural integrity, gait behavior and pressure points, supporting design decisions intended to improve prosthetic fit, comfort and performance.
- Directed project strategy, task allocation, technical decision-making and integration across CAD modelling, simulation, digital engineering and final solution presentation.
- Applied Siemens NX and Simcenter 3D to demonstrate how virtual engineering could support more efficient prosthetic design and testing.

Projects

- Wearable Lift Assist Arm Brace
- Virtual Prosthetic Test Lab
- Hand Actuation For Rehabilitation and Performance (HARP)
- High-Intensity Phototherapy Device
- Sleep Apnea Monitor
- Ezytherm
- Neonatal Incubator
- Fetal Heart Rate and Kick Counting Device
- Automated Neonatal Device (ANET)
- Design of an industrial Baking oven

SKILLS

- CAD / Product Design: SolidWorks | Siemens NX | 3D Modelling | Product Development | Design for Manufacturability | Prototyping
- Simulation / Analysis: Simcenter 3D | CAE | CFD | Virtual Testing | Engineering Simulation
- Electronics / Hardware: KiCad | Siemens Electronic Design Automation | Circuit Design | Embedded Systems
- Programming / Computational: Python | C++ | MATLAB
- Emerging Technologies: Artificial Intelligence | AI Assisted Engineering | Digital Twins | Digital Engineering | Engineering Automation

CERTIFICATIONS AND AWARDS

- SIEMENS 2026 EDA Tech Trek — Winner
- SIEMENS 2026 Immersive Design Challenge — Semifinalist
- Future Workforce-Rising star (SIEMENS)
- Project Management & Integrated Engineering Course (SIEMENS)

VOLUNTARY/EXTRA-CURRICULAR ACTIVITY

Helping Hands Foundation (Co-founder)

- Co-founded a community initiative focused on supporting underprivileged children.
- Organize annual donation drives supporting orphanages with essential supplies.
- Coordinated a 2024 and 2025 outreach that provided food, clothing and toiletries to Holy Child Orphanage.

REFEREES

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